

COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY

SCHOOL OF ENGINEERING

B. TECH DEGREE 1ST SEMESTER SECOND INTERNAL EXAMINATIONS, DECEMBER 2023

23-200-0101B: CALCULUS, Branch: ECE, Batch: A

(2023 Scheme)

TIME: 2hrs

MAX. MARKS: 30

Course Outcomes: On completion of this course the student will be able to

CO3	Evaluate area as double integrals and volume as triple integrals.
CO4	Illustrate the application and physical meaning of gradient, divergence and curl.

PART A (Answer all questions)

(5 x 2 = 10 Marks)

	Questions	Marks	BL	CO	PO
I (a)	If $\phi(x, y, z) = 3x^2y - y^3z^2$, find $\text{grad } \phi$ at the point (1,2,1)	2	1	4	1
(b)	If $r = \vec{r} $, where $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, find $\text{curl } \vec{r}$	2	1	4	1
(c)	Show that $\vec{F} = (x + 3y)\hat{i} + (y - 2z)\hat{j} + (x - 2z)\hat{k}$ is solenoidal.	2	2	4	1
(d)	Find the value of a if $\vec{F} = (axy - z^2)\hat{i} + (x^2 + ayz)\hat{j} + (y^2 - axz)\hat{k}$ is irrotational.	2	2	4	1
(e)	Compute the divergence of $\vec{F} = xyz\hat{i} + xyz\hat{j} + xyz\hat{k}$ at (1,1,1).	2	1	4	1

PART B (Answer the following questions choosing either the first OR second option from each module)

MODULE 4

(1 x 10 = 10 Marks)

Q.No.	Questions	Marks	BL	CO	PO
II (a)	If $r = \vec{r} $, where $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, Prove that $\nabla r^n = nr^{n-2}\vec{r}$.	5	3	4	1
(b)	Find the unit normal vector to the surface $x^2 + 2y^2 + z^2 = 10$ at (1,2,1)	5	4	4	1
OR					
III (a)	Evaluate $\int_C \vec{F} \cdot d\vec{r}$ if $\vec{F} = 3xy\hat{i} - y^2\hat{j}$ where C is the curve in the XY plane given by $y = x^3$ from (0,0) to (1,1).	5	5	3	1
(b)	Evaluate $\iiint_S \vec{F} \cdot \hat{n} \, ds$ using Gauss divergence theorem if S is the surface bounded by $x = 1, y = 1, z = 1, x = 2, y = 2, z = 2, \vec{F} = (x + y)\hat{i} + x\hat{j} + z\hat{k}$	5	4	3	1
MODULE 4 (1 x 10 = 10 Marks)					
IV	(a) Show that $\vec{F} = (6xy + z^2)\hat{i} + (3x^2 - z)\hat{j} + (3xz^2 - y)\hat{k}$ is irrotational	5	3	4	1
	(b) Find the potential function of $\vec{F}$.	5	5	4	1
OR					
V (a)	Evaluate using Stoke's theorem $\oint_C \vec{F} \cdot d\vec{r}$ where C is a curve in XY plane around the lines $x = 0, y = 0, x = 5, y = 5$ and $\vec{F} = x^2\hat{i} + xy\hat{j}$.	5	6	3	1
(b)	Evaluate $\iiint_V \vec{r} \, dv$, where $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ where V is the unit cube.	5	4	3	1

Bloom's Taxonomy Levels

L1 - %	L2 - %	L3 - %	L4 - %	L5 - %	L6 - %
23	15.4	15.4	23	15.4	7.5